

✓ LoRA MLP + Attention (Low Rank)

This notebook mirrors the full-run LoRA pipeline and focuses on low-rank target-module ablations for attention-only vs attention+MLP adapters.

```
from pathlib import Path
import sys
import pandas as pd
import torch
from IPython.display import display
```

✓ Colab setup and project root

```
try:
    from google.colab import drive
    IN_COLAB = True
except ImportError:
    IN_COLAB = False

if IN_COLAB:
    drive.mount('/content/drive')
    CANDIDATE_ROOT = Path('/content/drive/MyDrive/DL_Final_Project')
    PROJECT_ROOT = CANDIDATE_ROOT if CANDIDATE_ROOT.exists() else Path.
else:
    cwd = Path.cwd().resolve()
    if cwd.name == 'notebooks':
        PROJECT_ROOT = cwd.parent
    elif cwd.name == 'LoRA_study' and cwd.parent.name == 'notebooks':
        PROJECT_ROOT = cwd.parent.parent
    else:
        PROJECT_ROOT = cwd

print('IN_COLAB:', IN_COLAB)
print('PROJECT_ROOT:', PROJECT_ROOT)
if str(PROJECT_ROOT) not in sys.path:
    sys.path.append(str(PROJECT_ROOT))
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount
IN_COLAB: True
PROJECT_ROOT: /content/drive/MyDrive/DL_Final_Project
```

```

import importlib.metadata as md
import importlib.util
import subprocess

def _parse_version_tuple(version_str: str) -> tuple[int, ...]:
    parts = []
    for token in version_str.split('.'):
        digits = ''.join(ch for ch in token if ch.isdigit())
        parts.append(int(digits) if digits else 0)
    return tuple(parts)

required_torchao = (0, 16, 0)
if importlib.util.find_spec('torchao') is not None:
    v_str = md.version('torchao')
    print(f'Found torchao=={v_str}')
    if _parse_version_tuple(v_str) < required_torchao:
        print('Uninstalling incompatible torchao...')
        subprocess.check_call([sys.executable, '-m', 'pip', 'uninstall',
                                'torchao'])
        raise SystemExit('Removed incompatible torchao. Restart runtime')
    print('torchao is compatible.')
else:
    print('torchao is not installed (this is fine).')

```

```
torchao is not installed (this is fine).
```

```

from src.gen_lora_study_utils import (
    set_seed,
    load_split,
    apply_sanity_subset,
    cap_validation_rows,
)
from src.lora_best_candidates_full_run_utils import build_configs_from
from src.lora_tuning_block_utils import count_trainable_params_for_config

```

✓ Runtime controls (same style as full-run notebook)

```

save = True

# For the sweep, keep this False so you do not run test inference for

```

```

# After choosing the best validation config, rerun only that config w
generate_submission = False

# Empty means "all ablations" (same behavior as submission_ablation_id
submission_ablation_ids = ['score_lora_sft', 'score_lora_option_scori
save_val_predictions = True
val_prediction_ablation_ids = ['score_lora_sft', 'score_lora_option_s

sanity_check = True
n = 512 # ignored when sanity_check=False

seed = 42
max_val_examples = 256 #None
top_k = 5
batch_size = 4
clear_cuda_between_runs = True

# For option_scoring-only runs, loss is a smoother early-stopping sign
# For mixed option_scoring + SFT runs, use 'accuracy' instead; see no
early_stopping = None
epochs_per_val_check = None
early_stopping_metric = 'loss'
early_stopping_val_examples = 128 #None

DATA_DIR = PROJECT_ROOT / 'data'
IMAGE_DIR = DATA_DIR / 'images' / 'images'
OUT_DIR = PROJECT_ROOT / 'outputs' / '01_lora_sft_vs_option_scoring'
ADAPTER_DIR = OUT_DIR / 'adapters'
MODEL_ID = 'HuggingFaceTB/SmolVLM-500M-Instruct'
DEVICE = 'cuda' if torch.cuda.is_available() else 'cpu'

torch_dtype = (
    'bfloat16'
    if torch.cuda.is_available() and torch.cuda.is_bf16_supported()
    else ('float16' if torch.cuda.is_available() else 'auto')
)

print('CUDA available:', torch.cuda.is_available())
if torch.cuda.is_available():
    print('GPU:', torch.cuda.get_device_name(0))
print({
    'save': save,
    'generate_submission': generate_submission,
    'submission_ablation_ids': submission_ablation_ids,
    'save_val_predictions': save_val_predictions,

```

```

        'val_prediction_ablation_ids': val_prediction_ablation_ids,
        'sanity_check': sanity_check,
        'n': n,
        'device': DEVICE,
    })
    set_seed(seed)

```

```

CUDA available: True
GPU: NVIDIA A100-SXM4-80GB
{'save': True, 'generate_submission': False, 'submission_ablation_ids'

```

```

train_df = load_split(DATA_DIR, 'train')
val_df = load_split(DATA_DIR, 'val')
test_df = load_split(DATA_DIR, 'test')

train_df = apply_sanity_subset(train_df, sanity_check, n, seed)
val_df = apply_sanity_subset(val_df, sanity_check, n, seed)
test_df = apply_sanity_subset(test_df, sanity_check, 1008, seed)
val_df = cap_validation_rows(val_df, max_val_examples)

print('train rows used:', len(train_df))
print('validation rows used:', len(val_df))
print('test rows used:', len(test_df))
print('IMAGE_DIR exists:', IMAGE_DIR.exists())

```

```

train rows used: 512
validation rows used: 256
test rows used: 1008
IMAGE_DIR exists: True

```

✓ Candidate ablation spaces

The key differences are low-rank settings and target modules. Input/data ablations are toggled directly in each candidate config.

```

# Single best-attempt config after the SFT masking / generation / opti

input_ablation = {
    # The strategy note says the original images are small and recomm
    # Use 512 for this run. If you later find many images are non-squ
    # and control processor resolution instead.
    'image_size': 384,

```

```

# Keep augmentation light. These images likely contain diagrams/tables
'enable_image_augmentation': True,
'brightness_jitter': 0.03,
'slight_rotation_deg': 0.5,

# Use metadata because subject/grade/topic may help route the reader
'include_metadata_in_prompt': True,
'metadata_fields': ['subject', 'grade', 'topic'],
'metadata_header': 'Metadata',

# With output_format='letter_only', this is mostly harmless/unused
# but keep it simple in case other code references it.
'answer_prefix_text': 'Answer:',
'question_label': 'Question:',
'choices_label': 'Choices:',
'instruction_prefix': '',
}

prompt_structure = 'question_first'
context_mode = 'hint_lecture'
choice_format = 'letter_dot'

# Important: use letter-only with option scoring.
# This makes each candidate just "A", "B", "C", etc.
output_format = 'letter_only'

# Option scoring does not rely on free generation, but keep these same
max_new_tokens = 4
decoding_strategy = 'beam'
num_beams = 1
parse_rule = 'strict_first_letter'

# Option scoring is slower than SFT because it scores every candidate
# Start with 12 rather than 25; early stopping can still stop sooner.
epochs = 3

ablation_space = {
    'lora_sft': {
        # Main change: use multiple-choice candidate scoring rather than
        'training_objective': 'option_scoring',

        'prompt_structure': prompt_structure,
        'context_mode': context_mode,
    },

```

```

'choice_format': choice_format,
'output_format': output_format,
'max_new_tokens': max_new_tokens,
'decoding_strategy': decoding_strategy,
'num_beams': num_beams,
'parse_rule': parse_rule,

# DoRA + attention/MLP targets.
# r=4 should stay safely below your 5M trainable-parameter cap
# attention behavior and internal MLP representations.
'lora_r':8,
'lora_alpha': 16,
'lora_dropout': 0.03,
'target_modules': [
    'q_proj', 'k_proj', 'v_proj', 'o_proj',
],
'lora_bias': 'none',
'lora_task_type': 'CAUSAL_LM',
'use_dora': False,
'use_rslora': False,

'epochs': epochs,

# Your current option-scoring trainer is effectively row-by-row
# may not be used unless you patched it in. Keep this at 1.
'gradient_accumulation_steps': 1,

# DoRA usually benefits from a slightly lower LR than vanilla
'lr': 5e-5,
'weight_decay': 0.01,
'max_grad_norm': 1.0,

# These only matter if you added scheduler support.
# If not, they will remain harmless config fields.
'warmup_ratio': 0.05,
'scheduler_type': 'cosine',

**input_ablation,
},
'lora_option_scoring': {
    # Main change: use multiple-choice candidate scoring rather than
    'training_objective': 'sft_next_token',

    'prompt_structure': prompt_structure,
    'context_mode': context_mode,

```

```

        'choice_format': choice_format,
        'output_format': output_format,
        'max_new_tokens': max_new_tokens,
        'decoding_strategy': decoding_strategy,
        'num_beams': num_beams,
        'parse_rule': parse_rule,

        # DoRA + attention/MLP targets.
        # r=4 should stay safely below your 5M trainable-parameter cap
        # attention behavior and internal MLP representations.
        'lora_r': 8,
        'lora_alpha': 16,
        'lora_dropout': 0.03,
        'target_modules': [
            'q_proj', 'k_proj', 'v_proj', 'o_proj',
        ],
        'lora_bias': 'none',
        'lora_task_type': 'CAUSAL_LM',
        'use_dora': False,
        'use_rslora': False,

        'epochs': epochs,

        # Your current option-scoring trainer is effectively row-by-row
        # may not be used unless you patched it in. Keep this at 1.
        'gradient_accumulation_steps': 1,

        # DoRA usually benefits from a slightly lower LR than vanilla
        'lr': 5e-5,
        'weight_decay': 0.01,
        'max_grad_norm': 1.0,

        # These only matter if you added scheduler support.
        # If not, they will remain harmless config fields.
        'warmup_ratio': 0.05,
        'scheduler_type': 'cosine',

        **input_ablation,
    },
}

all_configs = build_configs_from_named_space(ablation_space, 'score')
display(pd.DataFrame(all_configs))

```

training objective prompt structure context mode choice format on

0	option_scoring	question_first	hint_lecture	letter_dot
1	sft_next_token	question_first	hint_lecture	letter_dot

2 rows x 37 columns

```
fixed_context = {
    'MODEL_ID': MODEL_ID,
    'DEVICE': DEVICE,
    'torch_dtype': torch_dtype,
    'IMAGE_DIR': IMAGE_DIR,
    'train_df': train_df,
    'val_df': val_df,
    'test_df': test_df,
    'batch_size': batch_size,
    'clear_cuda_between_runs': clear_cuda_between_runs,
    'early_stopping': early_stopping,
    'epochs_per_val_check': epochs_per_val_check,
    'early_stopping_metric': early_stopping_metric,
    'early_stopping_val_examples': early_stopping_val_examples,
    # Save adapters once during val training so test submissions can
    'save_trained_adapters': bool(generate_submission),
    'ADAPTER_DIR': ADAPTER_DIR,
}

_param_rows = []
for cfg in all_configs:
    counts = count_trainable_params_for_config(cfg, {'MODEL_ID': MODEL_ID})
    _param_rows.append({
        'ablation_id': cfg['ablation_id'],
        'trainable_params': counts['trainable'],
        'total_params': counts['total'],
        'trainable_pct': round(float(counts['trainable_pct']), 4),
    })
display(pd.DataFrame(_param_rows).sort_values('ablation_id').reset_index())
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
warnings.warn(
```

Warning: You are sending unauthenticated requests to the HF Hub. Please
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please
 `torch\_dtype` is deprecated! Use `dtype` instead!

Loading weights: 100% 489/489 [00:00<00:00, 1142.50it/s, Materializing param=model.vision_model.

Loading weights: 100% 489/489 [00:00<00:00, 1168.81it/s, Materializing param=model.vision_model.

	ablation_id	trainable_params	total_params	trainable_pct
0	score_lora_option_scoring	2080768	509563072	0.4083
1	score_lora_sft	2080768	509563072	0.4083

```

results = []
val_preds = []
histories = []
submission_frames = []

config_by_id = {cfg['ablation_id']: cfg for cfg in all_configs}
all_ablation_ids = set(config_by_id.keys())

submission_requested_ids = [str(x) for x in (submission_ablation_ids -
submission_selected_ids = set(submission_requested_ids) if submission_
submission_missing_ids = sorted(set(submission_requested_ids) - all_ablation_ids)
if submission_missing_ids:
    print('Skipping unknown submission ablation ids:', submission_missing_ids)

val_prediction_requested_ids = [str(x) for x in (val_prediction_ablation_ids -
val_prediction_selected_ids = set(val_prediction_requested_ids) if val_prediction_
val_prediction_missing_ids = sorted(set(val_prediction_requested_ids) - all_val_prediction_ids)
if val_prediction_missing_ids:
    print('Skipping unknown validation-prediction ablation ids:', val_prediction_missing_ids)

for cfg in all_configs:
    ablation_id = cfg['ablation_id']
    summary, val_pred_df, history_df = run_lora_candidate_on_val(cfg)
    results.append(summary)
    val_preds.append(val_pred_df)
    if len(history_df):
        histories.append(history_df)

# Generate submission immediately after training for selected ablation ids
if generate_submission and (ablation_id in submission_selected_ids):
    adapter_path = ADAPTER_DIR / ablation_id

```

```

test_pred_df, submission_df = run_test_submission_from_saved(
    config=cfg,
    fixed_context=fixed_context,
    adapter_dir=adapter_path,
    save=save,
    out_dir=OUT_DIR,
    filename_prefix=f'mlp_attn_{ablation_id}_',
)
_sub = submission_df.copy()
_sub['ablation_id'] = ablation_id
submission_frames.append(_sub)
print(f'Generated submission without retraining for: {ablation_id}')
display(test_pred_df.head())
display(submission_df.head())

if clear_cuda_between_runs and torch.cuda.is_available():
    torch.cuda.empty_cache()

results_df = pd.DataFrame(results).sort_values('accuracy', ascending=False)
val_predictions_df = pd.concat(val_preds, ignore_index=True) if val_preds else pd.DataFrame()
history_df = pd.concat(histories, ignore_index=True) if histories else pd.DataFrame()
display(results_df)

if submission_frames:
    combined_submission_preview = pd.concat(submission_frames, ignore_index=True)
    display(combined_submission_preview.head())

if save and save_val_predictions:
    if val_prediction_requested_ids:
        selected_val_predictions_df = val_predictions_df[
            val_predictions_df['ablation_id'].isin(sorted(val_prediction_requested_ids))
        ].reset_index(drop=True)
    else:
        selected_val_predictions_df = val_predictions_df.copy()

    if len(selected_val_predictions_df):
        OUT_DIR.mkdir(parents=True, exist_ok=True)
        val_pred_path = OUT_DIR / 'mlp_attn_val_predictions.csv'
        selected_val_predictions_df.to_csv(val_pred_path, index=False)
        print(
            'Saved validation predictions:',
            {
                'path': str(val_pred_path),
                'rows': int(len(selected_val_predictions_df)),
                'ablations': sorted(selected_val_predictions_df['ablation_id'])
            }
        )

```

```

        },
    )
    display(selected_val_predictions_df.head())
else:
    print('No validation predictions matched the selected ablation')
elif save_val_predictions and not save:
    print('save_val_predictions=True but save=False; skipping validation')

```

Loading weights: 100% 489/489 [00:00<00:00, 1174.15it/s, Materializing param=model.vision_model.

Score epoch 1/3: 100% 512/512 [25:17<00:00, 3.04s/it]

Score epoch 2/3: 100% 512/512 [20:34<00:00, 2.36s/it]

Score epoch 3/3: 100% 512/512 [20:34<00:00, 2.36s/it]

Val option scoring: 100% 256/256 [11:51<00:00, 2.77s/it]

Loading weights: 100% 489/489 [00:00<00:00, 1194.21it/s, Materializing param=model.vision_model.

SFT epoch 1/3: 100% 128/128 [09:04<00:00, 4.24s/it]

SFT epoch 2/3: 100% 128/128 [09:03<00:00, 4.22s/it]

SFT epoch 3/3: 100% 128/128 [09:01<00:00, 4.22s/it]

Val teacher- 64/64 [04:01<00:00, 3.75s/it]

forced loss: 100%

	training_objective	prompt_structure	context_mode	choice_format	on
0	option_scoring	question_first	hint_lecture	letter_dot	
1	sft_next_token	question_first	hint_lecture	letter_dot	

2 rows x 45 columns

Saved validation predictions: {'path': '/content/drive/MyDrive/DL_Final

	id	pred_answer	num_choices	gold_answer	is_correct	ablation
0	val_00570	2.0	3.0	0.0	0.0	score_lora
1	val_03508	2.0	3.0	2.0	1.0	score_lora
2	val_03587	0.0	3.0	0.0	1.0	score_lora
3	val_00603	3.0	4.0	3.0	1.0	score_lora

Start coding or generate with AI.